

# Programmatoid and neuronoid computations

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June 26, 2026

What about ?

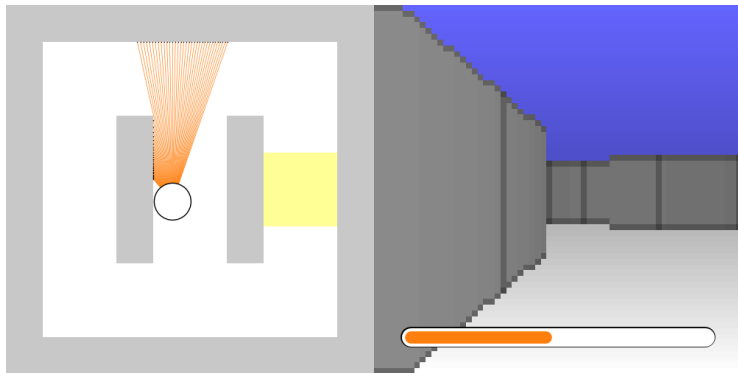
# *Programming* (not training) neural networks

- ▶ With a benchmark and ...
- ▶ Using computer algebra.

## But ... why ?

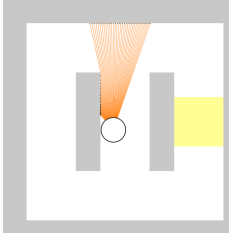
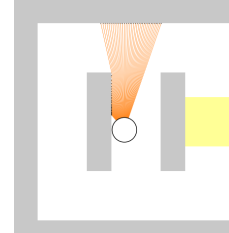
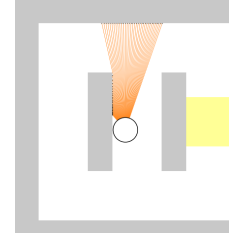
- ▶ For the fun :)
- ▶ To illustrate a feed-forward neural network computational property.
- ▶ To provide an approximate neural network complexity lower bound, given a task.
- ▶ To obtain a parsimonious computation tool.
- ▶ Why not ?

# The braincraft benchmark

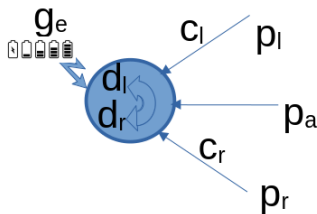


Documented in the README ☒ (please).

## Three tasks:

|                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                       |                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>The diagram shows a robot (a small white circle) in the center of a corridor. To the left and right of the robot are two gray vertical bars. To the right of the right bar is a yellow rectangular area representing an energy source. Orange concentric lines radiate from the robot, representing its field of view or sensor range.</p> |  <p>The diagram is identical to the first one, showing a robot in a corridor with an energy source to the right.</p> |  <p>The diagram is identical to the first one, showing a robot in a corridor with an energy source to the right.</p> |
| <p>Find the energy source leftward or rightward ✓.</p>                                                                                                                                                                                                                                                                                                                                                                          | <p>Find the correct direction, given the blue cue ✓.</p>                                                                                                                                              | <p>Choose the best energy source ✓.</p>                                                                                                                                                                |

## A (simplified) bot, with:

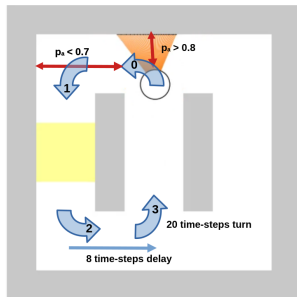
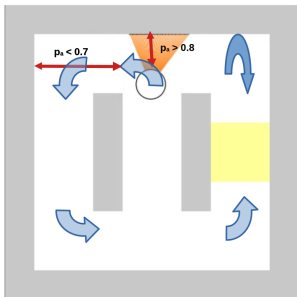


- left, ahead and right proximity sensors

$$p_{\bullet} \in ]0_{\text{no-wall}}, 1_{\text{wall-hit}}], \bullet \in \{l, a, r\},$$

- color sensors  $c_{\bullet \dagger}$ ,
- energy gauge  $g_e \in [0_{\text{death}}, 1_{\text{full}}]$ ,
- running at constant velocity, and
- turning leftward or rightward  $d_l \in [0, 1_{+5^\circ}]$ .

# Heuristics:



- ▶ Stabilizes the forward motion using linear feedback correction.
- ▶ Performs quarter-turn using ahead proximity thresholds.
- ▶ Performs about-turn or half-corridor turn using delays.
- ▶ Stores the preferred quarter-turn direction, e.g. from cue.
- ▶ Stores previous energy feeds for comparison.